

## **ECA0303-SIGNALS AND SYSTEM**

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### **Experiment 1.2**

Generation of Discrete Time Signals

#### **Program:**

% Generation of Discrete Time Signal

clc;

clear;

close all;

% Discrete-time index

n = -10:10;

% Discrete-time sinusoidal signal

x = sin(0.2\*pi\*n);

% Plotting

figure;

stem(n, x, 'filled');

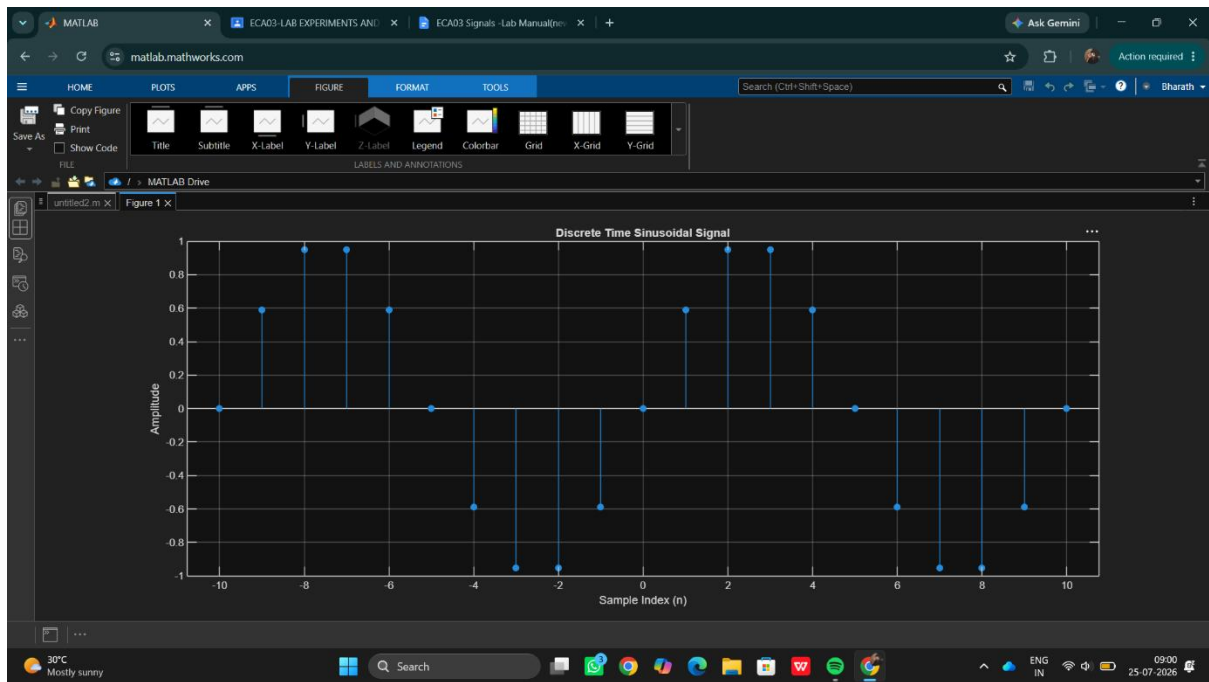
grid on;

title('Discrete Time Sinusoidal Signal');

xlabel('Sample Index (n)');

ylabel('Amplitude');

#### **Output Waveform:**



## Result:

This test case validates that the signal generator can accurately produce a discrete-time unit impulse signal at a specified index, which is essential for analyzing the response of discrete time systems.